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Combinatorial synthesis and analysis of $Al_xTa_yV_z$-$Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ and $Al_{10}CrMo_xNbTiZr_{10}$ refractory high-entropy alloys: Oxidation behavior

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ABSTRACT

The combinatorial development of refractory high-entropy alloy $Al_xTa_yV_z$-$Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ ($Al_xTa_yV_z$-Q) was carried out, and microstructural analysis was performed. The homogenized $Al_xTa_yV_z$-Q revealed a body-centered cubic structure with intermetallic phases. High-temperature oxidation analysis of $Al_xTa_yV_z$-Q for 1 h at 1000 °C using thermogravimetric analysis (TGA) revealed volatile oxidation of the alloy. Therefore, in an effort to improve the oxidation resistance of the alloy, the composition was modified to $Al_{10}CrMo_xNbTiZr_{10}$ and analyzed. The TGA analysis revealed enhanced oxidation resistance of $Al_{10}CrNbTiZr_{10}$ (Mo-0), and a weight gain of only 1 mg/cm^2 after oxidation for 1 h at 1000 °C in air, owing to the formation of the protective oxides of Al and Cr. The Mo-x samples were subjected to prolonged oxidation (for 50 h) at 1000 °C in air. After 50 h of oxidation, the Mo-0 sample showed a weight gain of ~24 mg/cm^2 and remained intact. The energy dispersive spectroscopy analysis of the oxide scale formed after 50 h of oxidation revealed $CrNbO_4$, Al_2O_3, and $AlTiO_5$, which account for the enhanced oxidation resistance of Mo-0 and forecasts its potential for high-temperature applications.

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1. Introduction

Owing to their high melting temperature and high-temperature mechanical properties, refractory metals and alloys are being developed for high-temperature structural materials [1]. However, refractory metals and alloys exhibit poor resistance to oxidation at high temperatures [2]. In order to improve the high-temperature oxidation resistance of refractory metals and alloys, various approaches have been taken [3], for example, adding various alloying elements (such as Al, Cr, and Si), which are able to form protective layers of oxides, including Al_2O_3, Cr_2O_3, and SiO_2 [4–7]. However, the addition of such elements forms brittle intermetallic compounds, such as Nb_3Al, $NbCr_2$, Nb_5Si_3, and $MoSi_2$, limiting possible applications because of the degradation of mechanical properties [8].

High-entropy alloys (HEAs) are a promising material, as they can overcome some of these barriers. HEAs consist of more than five metallic elements in equiatomic or near-equiatomic proportions [9–11]. The high configurational entropy of HEAs promotes solid-solution formation, which improves their high-temperature mechanical properties by increasing the resistance to softening at high temperatures [12–14]. Refractory HEAs (RHEAs) have gained particular interest, as they show superior mechanical properties at high temperatures as compared with nickel-based superalloys [15–18]. However, the poor oxidation resistance of RHEAs restricts their applications at elevated temperatures [19].

Several studies on the high-temperature oxidation behavior of RHEAs have been reported [19–28]. $CrMo_{0.5}NbTa_{0.5}TiZr$ exhibits parabolic oxidation at 1000 °C for 100 h; however, spalling was observed after oxidation [19]. $Al_{0.5}CrMoNbTi$, $Al_{0.5}CrMoNbV$, $Al_{0.5}CrMoNbTiV$, and $Al_{0.5}CrMoNbSi_{0.3}TiV$ alloys showed linear oxidation when oxidized at 1300 °C for 20 h [29]. The oxidation resistance of these alloys was found to improve with the addition of Ti and Si and the removal of V [29]. AlCrMoTaTi, when oxidized at 900–1100 °C for 48 h, exhibited parabolic oxidation and protective and dense alumina underneath a rutile layer [27]. The addition of 1 at.% Si further improved the oxidation resistance of AlCrMoNbTi [26]. AlCrMoTiW was oxidized at 1000 °C for 40 h and

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Abbreviations

Energy Dispersive Spectroscopy EDS
High-Entropy Alloys HEA
Scanning Electron Microscopy SEM
Refractory HEA RHEA
Thermogravimetric Analysis TGA
X-ray Diffraction XRD

inhomogeneous and porous oxide scale, which exhibited a tendency to spall, was observed [22]. NbTiVZr showed rapid and linear oxidation at 1000 °C. The replacement of V with Cr to form CrNbTiZr resulted in parabolic oxidation, owing to the formation of oxides composed of $CrNbO_4$ and ZrO_2 [23]. AlNbTiZr also exhibited parabolic oxidation when oxidized at 600–1000 °C for 50 h, with complex oxides $AlNbO_4$ and Ti_2ZrO_6 formed during oxidation at 1000 °C [30]. The oxidation behavior of $AlNbTiVZr_{0.25}$ was studied at 600–900 °C and a weight gain of 0.31–225 mg/cm^2 was obtained after 50 h of oxidation. Moreover, $AlNbTiVZr_{0.25}$ got disintegrated after oxidation at 900 °C for 50 h [31]. The addition of Al in $Al_xHfNbTaTiZr$ significantly improved its oxidation resistance at 700–1100 °C (for 100 h) [32]. To assess the oxidation resistance of metals and alloys, the weight gain during oxidation is measured. A summary of the weight gain of various RHEAs during oxidation at 1000 °C in an air environment is presented in Table 1.

The types of elements that make up the RHEAs and their concentrations have a significant effect on the properties of RHEAs, as seen in Table 1. Thus, a systemic development of new RHEAs is needed, so that combinatorial libraries of the properties of RHEAs can be developed and promising compositions can be identified for further research. Therefore, a combinatorial approach was employed to develop combinatorial libraries for the study of the oxidation resistance of $Al_xCr_yMo_zNbTiZr$ RHEAs [33]. The oxidation resistance analysis of $Al_xCr_yMo_zNbTiZr$ RHEAs at 1000 °C for 20 h revealed a comparatively reduced weight gain in $AlCr_{0.5}Mo_{0.5}NbTiZr$ (21 mg/cm^2) and $Al_{1.5}Cr_{0.5}NbTiZr$ (20 mg/cm^2) [33]. A combinatorial library of the oxidation resistance of $Al_xTa_yV_zCr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ (Al_x-Ta_yV_z-Q, where Q = $Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$, and x, y, z are in atomic percent) was developed. To improve the oxidation resistance further, the Zr content was reduced to 10 at.% [39], and Al, Ta, and V were varied to obtain the optimal combination of oxidation resistance and mechanical properties [40,41]. Fig. 1 depicts the combinatorial approach. Based on the results, $Al_{10}CrMoNbTiZr_{10}$ was selected and the effect of Mo analyzed by developing $Al_{10}CrMo_xNbTiZr_{10}$ (Mo-x), which revealed the promising oxidation resistant alloy, $Al_{10}CrNbTiZr_{10}$ (0 at.%Mo).

Table 1
Weight gain during oxidation of RHEAs at 1000 °C in air.

Alloy composition	Weight Gain (mg/cm^2)				Ref.
	1 h	3 h	20 h	50 h	
$CrMo_{0.5}NbTa_{0.5}TiZr$	55	110			[19]
CrNbTiZr	13	25	60	83	[23]
NbTiVZr	30	100			[23]
AlCrMoNbTi	0.5	0.8	3.5	9.0	[26]
AlCrMoNbTi–1Si	0.5	1	2.4	6.4	[26]
AlCrMoTiW	2.3	3.8	6.5		[22]
$AlCr_{0.5}Mo_{0.5}NbTiZr$	4.29	21			[33]
$AlCr_{0.5}Mo_{0.5}NbTiZr$	3.46				[33]
$Al_{0.5}CrMo_{0.5}NbTiZr$	13.38				[33]
$Al_{0.5}Cr_{0.5}MoNbTiZr$	7.29				[33]
$Al_{1.5}Mo_{0.5}NbTiZr$	8.25				[33]
AlMoNbTiZr	1.27				[33]
$Al_{0.5}Mo_{1.5}NbTiZr$	6.3	20			[33]
$Al_{1.5}Cr_{0.5}NbTiZr$	10.66				[33]
AlCrNbTiZr	6.1	39			[33]
$Al_{0.5}Cr_{1.5}NbTiZr$	9.4				[33]
$Cr_{1.5}Mo_{0.5}NbTiZr$	8.77				[33]
CrMoNbTiZr	0.5				[33]
$Cr_{0.5}Mo_{1.5}NbTiZr$	7.7		13.6		[30]
AlCrNbTiZr	13	13	23	52	[34]
AlCrMoTaTi	0.15	0.3	0.5	0.6	[35]
AlCrMoTaTi–1Si	0.25	0.4	0.8	1.3	[35]
HfNbTaTiZr	17				[32]
$Al_{0.5}HfNbTaTiZr$	8.5				[32]
AlHfNbTaTiZr	6				[32]
AlNbTiZr	3.8				[30]
HfNbTiZr	0.25				[30]
$AlCoCrFeMo_{0.5}NiSiTi$	0.3	1	2.2	3.7	[36]
$AlCrFeMo_{0.5}NiSiTi$	1	1.8	2.8	8	[36]
CrMoNbTaV	3	13	42		[37]
$AlNb_{1.5}Ta_{0.5}Ti_{1.5}Zr_{0.5}$	5	10	25	38	[38]

2. Experimental

$Al_xTa_yV_z$-Q RHEAs were prepared by melting 99.9% metal sources using an ACM-01 arc melting furnace (DAIA-VACUUM, Japan). All samples were remelted five times. In order to improve the chemical homogeneity further, the RHEA samples were heated to 1200 °C and held at this temperature for 24 h; then, air cooling was performed [42–44]. All samples were cut into identical shapes and sizes, and subjected to thermogravimetric analysis (TGA) for 1 h at 1000 °C in air.

After TGA analysis of $Al_xTa_yV_z$-Q RHEAs, the composition was modified and $Al_{10}CrMo_xNbTiZr_{10}$ RHEAs developed. Microstructural examinations were carried out by X-ray diffraction (XRD, D/MAX-2500, RIGAKU, USA), scanning electron microscopy (SEM, FEI Magellan 400, USA), and energy dispersive spectroscopy (EDS, coupled with SEM). The effect of the Mo concentration on the oxidation behavior of $Al_{10}Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ was analyzed by preparing $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ (with 0 at.%Mo, Mo-0), $Al_{12}Cr_{24}$-$Mo_4Nb_{24}Ti_{24}Zr_{12}$ (with 4 at.%Mo, Mo-4), and $Al_{11.5}Cr_{23}Mo_8Nb_{23}$-$Ti_{23}Zr_{11.5}$ (with 8 at.%Mo, Mo-4) RHEAs. The oxidation behavior was analyzed by TGA at 1000 °C for 1 h in an air atmosphere. Prolonged oxidation tests were conducted at 1000 °C for up to 50 h. Cubic alloy samples (with dimensions of 5 mm × 5 mm × 5 mm) were prepared for prolonged oxidation tests. A lab scale box furnace was used to probe the oxidation of the alloys in an air environment. At first, the temperature was increased to 1000 °C, after that HEA samples were loaded in the furnace. The change in weight during oxidation was measured up to 50 h with a time step of 10 h. The oxidized surfaces were characterized by XRD and SEM.

3. Results and analysis

Pellet-shaped $Al_xTa_yV_z$-Q samples of identical shape, size, and surface area, were subjected to TGA, and the weight change per unit surface area during oxidation at 1000 °C in air was analyzed. The change in mass during oxidation and the total mass change per unit area after 1 h of oxidation is shown in Fig. 2a and (b), respectively. Since it is difficult to distinguish the oxidation curves, the weight gain data is presented in Table S1 (supplementary material).

The weight gain of $Al_xTa_yV_z$-Q due to oxidation at 1000 °C for 1 h varies from 0.8 mg/cm^2 to 8.5 mg/cm^2. V_{10}-Q exhibits the highest weight gain (8.5 mg/cm^2) after 1 h of oxidation at 1000 °C. The reduction in the oxidation resistance of RHEAs due to the addition of V has been reported [29]. The oxidation of $Al_{0.5}CrMoNbTi$ at 1300 °C shows a weight gain of ~150 mg/cm^2 after 20 h [29],

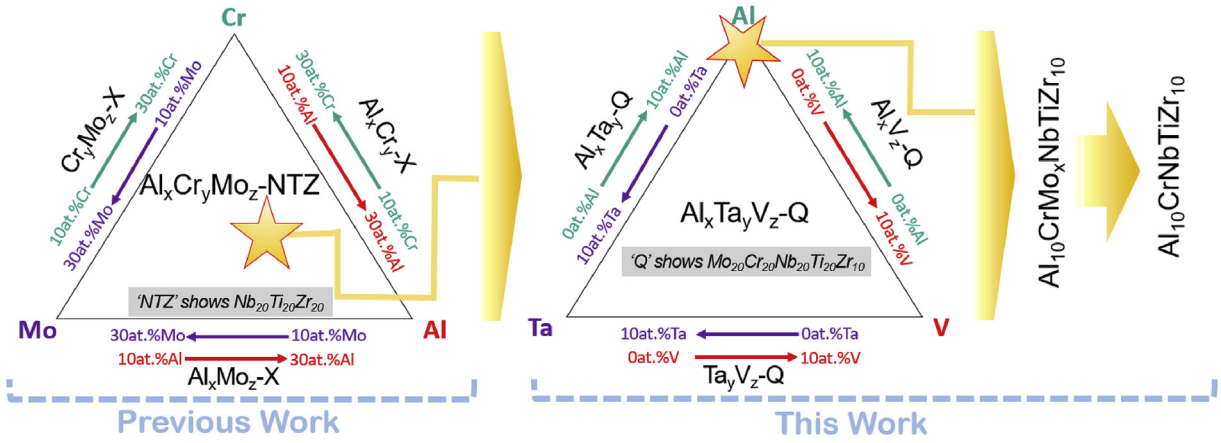


Fig. 1. A combinatorial approach for the development and analysis of the $Al_xTa_yV_zCr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ ($Al_xTa_yV_z$-Q, where Q = $Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$).

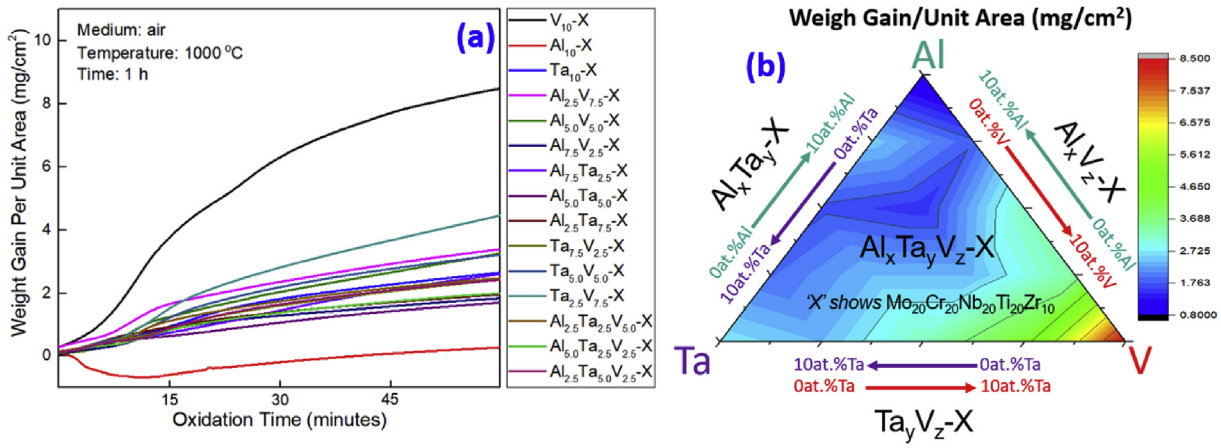


Fig. 2. (a) TGA curves and (b) total mass change/unit area after 1 h of oxidation at 1000 °C for homogenized $Al_xTa_yV_z$-Q.

whereas the addition of V to form $Al_{0.5}CrMoNbV$ increases this weight gain to ~275 mg/cm^2 [29]. The adverse effects of V on the oxidation resistance of RHEA are also seen from the oxidation resistance analysis of NbTiZrCr and NbTiZrV at 1000 °C. After 1 h of oxidation, NbTiZrCr shows a weight gain of 13 mg/cm^2, whereas the weight gain in NbTiZrV reaches 30 mg/cm^2 [23]. Al_xV_z-Q follows the same trend, showing an increase in weight gain from 1.8 mg/cm^2 to 3.4 mg/cm^2 with an increase in V. Although the oxide of Ta is not protective, it demonstrates strong bonding with the base metal [45]. Therefore, Ta_{10}-Q exhibits relatively less weight gain (2.7 mg/cm^2) as compared with the weight gain of V_{10}-Q. However, the unprotective nature of the oxides of Ta result in an increase in weight gain with an increase in Ta. Ta_yV_z-Q shows an increase in weight gain from 2.4 mg/cm^2 to 4.4 mg/cm^2 with Ta, because the oxides of Ta are not protective at high temperatures [46].

Al forms a protective oxide scale and improves the oxidation resistance of RHEAs. NbTiVZr shows a weight gain of 30 mg/cm^2 after 1 h of oxidation at 1000 °C [30], whereas AlNbTiVZr (containing Al, otherwise similar to NbTiVZr), when oxidized under the same conditions, shows a weight gain of 3.8 mg/cm^2 [30]. The oxidation resistance analysis of $Al_xHfNbTaTiZr$ at 1100 °C for 100 h in air also shows a reduction in weight gain from 77 mg/cm^2 to 50 mg/cm^2, owing to the increase in Al from 0 to 16.6 at.% [32]. Therefore, Al_{10}-Q shows reduced weight gain (0.8 mg/cm^2) after 1 h of oxidation at 1000 °C in air. However, the Al_{10}-Q HEA shows a negative slope for the weight gain curve for up to 10 min of oxidation. The negative slope of the TGA curve (during the first 10 min only) is attributed to volatile oxidation of Mo in the early stage of oxidation. After 10 min, the slope becomes positive. The oxidation of Mo at high temperature forms MoO_3, which volatilizes at temperatures above 700 °C [47,48]. The volatile oxidation of Mo-containing RHEAs [33] and Mo-rich refractory alloys [49] has also been reported. After 10 min of oxidation, the slope of the weight gain curve becomes positive, which can be explained as follows. After the evaporation of MoO_3, the Mo-depleted RHEA is exposed to oxygen and forms several other oxides. These oxides reduce the inward diffusion and reaction of oxygen with Mo by acting as a barrier [47]; hence, the TGA curve becomes positive. The $Al_xTa_yV_z$-Q RHEAs, other than Al_{10}-Q, show relatively high weight gain; therefore, the TGA curves do not exhibit negative slopes.

The XRD patterns and SEM microstructures of homogenized (1200 °C, 24 h, air cooling) Mo-0, Mo-4, and Mo-8 samples are shown in Fig. 3.

The XRD patterns consist mainly of a body-centered cubic (BCC) phase. The secondary phases (Zr_2Al, Cr_2Zr, and C15 Laves intermetallics) are also seen in the XRD patterns. The SEM microstructure (Fig. 3b) shows a dual-phase microstructure. The chemical maps (Fig. 3c) of Mo-0 show Nb- and Ti-rich solid-solution phases, whereas Al, Zr, and Cr are enriched in the secondary phase (Laves intermetallics). The chemical maps also show the

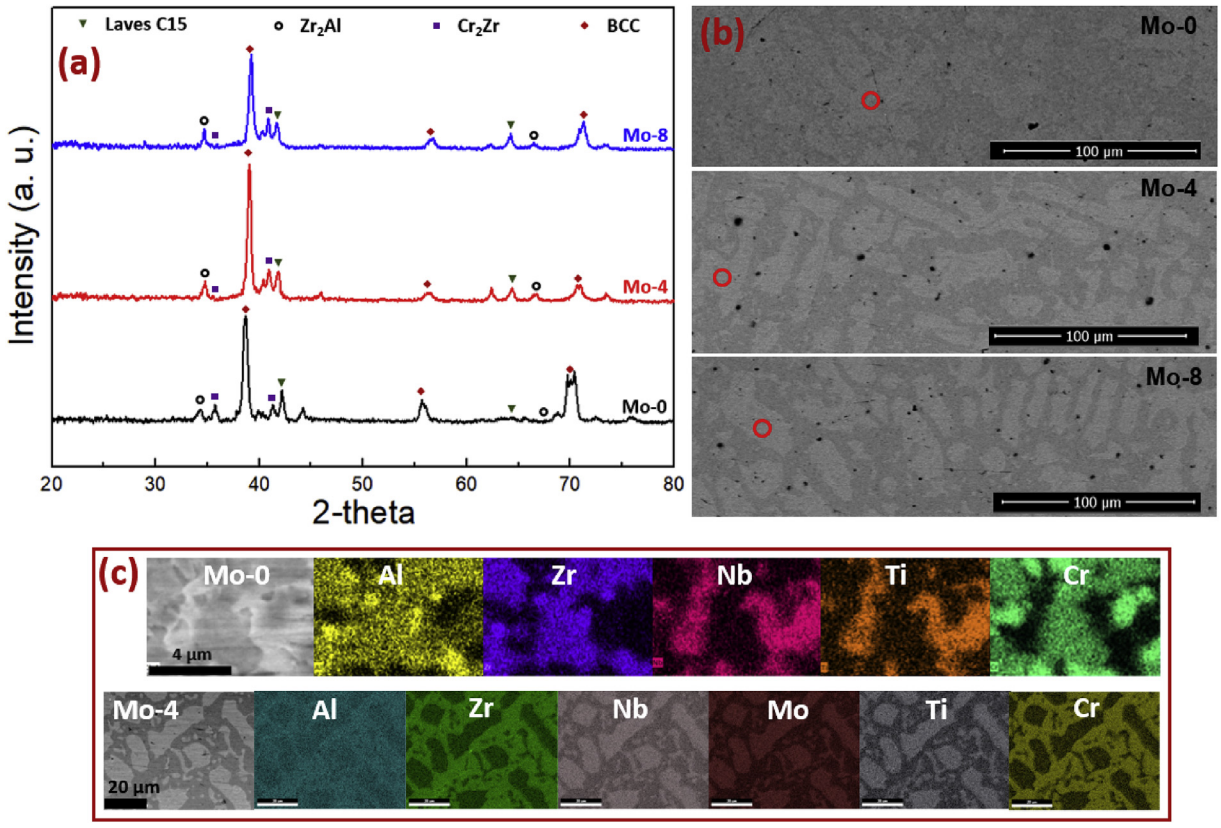


Fig. 3. (a) XRD, (b) SEM, and (c) EDS analysis of homogenized Mo-0, Mo-4, and Mo-8.

presence of Mo (in Mo-4) in the solid solution phases of the respective RHEAs. The addition of Mo did not change the composition of the Laves intermetallic phases, as they remain enriched in Al, Zr, and Cr. The point EDS analysis of various phases observed under SEM is shown in Table 2.

The TGA analysis of Mo-0, Mo-4, and Mo-8 is shown in Fig. 4a. The TGA curves of all samples show a positive slope and parabolic nature, which indicates that the evaporation (as observed in the TGA analysis of Al_{10}-Q (Fig. 2a) has been successfully suppressed. Mo-0 shows a weight gain of 1 mg/cm^2 after 1 h of oxidation at 1000 °C in air. As the Mo content increases to 8 at.%, the weight gain after oxidation becomes ~3 mg/cm^2. The thickness of the oxide layer shows the same trend. As the Mo content increases from 0 at.% to 8 at.%, the thickness of the oxide layer increases from ~20 μm to ~51 μm. Meanwhile, the oxygen-penetrated layer remains in the range of ~60 μm—~70 μm. The samples having a higher Mo concentration (i.e. Mo-12 and Mo-16) were also analyzed, and a decrease in weight gain (compared to that of Mo-8) during oxidation is observed, which suggests the evaporation of Mo oxides.

The oxide scale on RHEAs show a mixture of oxides [19,33,50]. Similar to previous studies, the XRD and EDS analysis (Fig. 4c and d, respectively) of the oxide scales on Mo-0, Mo-4, and Mo-8 also show the presence of a mixture of various oxides. Although the primary XRD peaks from the oxide scale on the Mo-0 sample match Nb_2O_5, the TEM-EDS point analysis of the regions relatively enriched in Nb show only 27.5—32.6 at.% Nb, 33.7—36.8 at.% O, and 3.0—8.7 at.% Zr, but the rest of the oxide scale (*i.e.* 21.9—35.8 at.%) contains Al, Ti and Cr, which form protective oxide scales. Discontinuous oxides of Al, containing other elements, are also observed via TEM-EDS. Zhang et al. recently reported an oxidation resistance analysis of a similar RHEA (AlCrNbTiZr) with the same constituents as in the developed Mo-0 RHEA [34]. They also observed a mixture of oxides in the oxide scale that developed during oxidation at 1000 °C [34], along with discontinuous Al_2O_3 at the outermost surface [34]. Zhang et al. reported an ~18 μm-thick oxide layer after 5 h of oxidation. Due to the relatively smaller concentration of Al in the presented samples, no continuous Al_2O_3 is observed in this study, and the thickness of the oxide layer reaches ~20 μm after 1 h of oxidation. The presence of Zr increases the diffusion of oxygen through the metal layer [39,45], as AlCrNbTiZr contains a higher Zr content [34] as compared with Mo-0. Therefore, AlCrNbTiZr shows a relatively higher thickness (~110 μm) for the oxygen-penetrated layer.

Owing to the promising behavior (in terms of weight gain during oxidation) of Mo-0, as compared with that of Al_{10}-Q and several

Table 2
The point EDS analysis of Mo-0, Mo-4, and Mo-8.

Sample name	Phases	Composition (at.%)					
		Al	Cr	Nb	Mo	Ti	Zr
Mo-0	BCC, 40 vol%	14.8	8.3	33.7	–	36.0	7.2
	Laves, 60 vol%	19.0	36.7	11.4	–	13.8	19.2
Mo-4	BCC, 60 vol%	13.6	10.2	32.7	4.5	33.3	5.7
	Laves, 40 vol%	17.3	38.3	12.5	1.9	12.2	17.9
Mo-8	BCC, 60 vol%	13.1	11.8	30.0	7.8	32.0	5.3
	Laves, 40 vol%	17.8	36.2	12.1	3.8	11.8	18.4

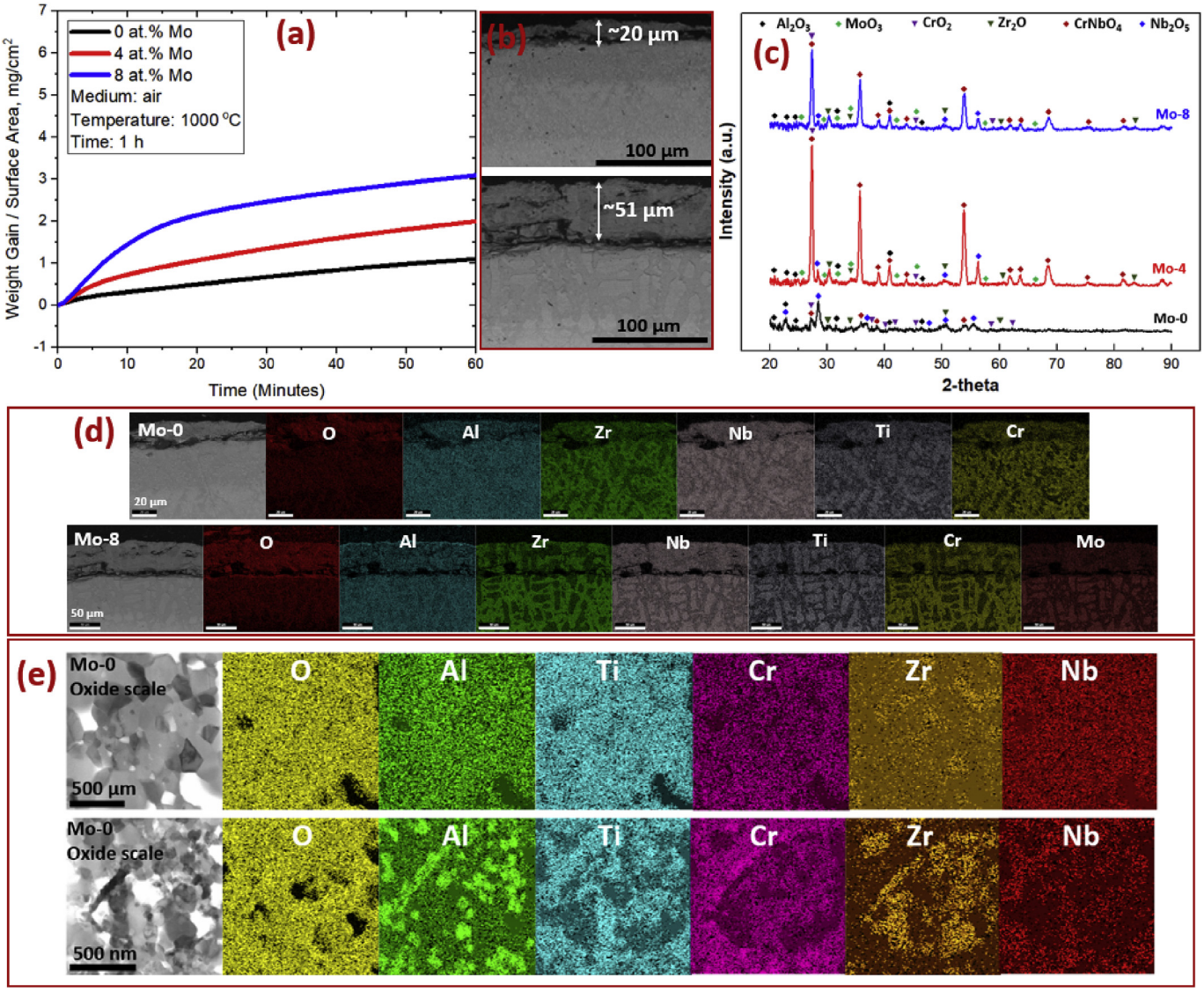


Fig. 4. (a) TGA curves, (b) SEM, (c) XRD, (d) EDS of oxidized Mo-0, Mo-4, and Mo-8 RHEAs and (e) TEM-EDS of oxide scale on Mo-0.

other RHEAs listed in Table 1, the weight gain after prolonged oxidation at high temperature (50 h, 1000 °C, air) is examined. The results are shown in Fig. 5.

Cubical samples with a dimension of 5 mm × 5 mm × 5 mm are used for the prolonged oxidation, as shown in Fig. 5a (i). As the oxidation progresses, the sample starts to gain weight. The sample oxidized for 20 h (Fig. 5a (ii)), shows oxide scale on the surface; however, no spallation or cracks are observed. After 30 h of oxidation, the oxide scale chips off from the surface, as seen in Fig. 5a (iii). After 50 h of oxidation, the Mo-0 sample shows a weight gain of ~24 mg/cm^2 (Fig. 5a (v)). Unlike $Hf_{0.5}Nb_{0.5}Ta_{0.5}Ti_{1.5}Zr$ [46] and HfTiZrNbTa [32], Mo-0 does not exhibit pesting. For comparison, the images of the Mo-4 and Mo-8 samples, oxidized for 50 h are also shown. These samples start to disintegrate after 10 h of oxidation, and after 50 h of oxidation, these samples are a fragmented mass, as shown in Fig. 5b. Owing to the change in the surface area as a result of disintegration, the weight gain analysis of the Mo-4 and Mo-8 was left undone. This sort of pesting/disintegration can be avoided by alloying (the gist of the present study) and surface coating [28].

The XRD analysis after oxidation for 50 h is shown in Fig. 5c. The XRD pattern shows that the oxide scale mainly consists of $CrNbO_4$ [34], which accounts for the promising oxidation resistance [23,34]. In addition to $CrNbO_4$, the oxides of Al and Ti also show clear peaks, which suggests the formation of protective oxides. The oxidation resistance of Mo-0 is higher than AlCrNbTiZr as well [34], which is attributed to the relatively lower Zr content in the nominal composition of Mo-0. ZrO_2 does not prevent oxygen diffusion into the base alloy, and the presence of Zr reduces the oxidation resistance [23]. The interatomic bonding of the oxides of Zr with the base metal gets weaker at higher temperatures, and the diffusion of metal and oxygen through the metal layer increases, which accelerates the growth of oxides [39,45]. Therefore, Mo-0 shows superior oxidation resistance at 1000 °C.

Owing to the poor oxidation resistance of RHEAs, their commercial applications are restricted [19]. In order to overcome this issue, the combinatorial synthesis of $Al_xTa_yV_zCr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ ($Al_xTa_yV_z$-Q) and $Al_{10}CrMo_xNbTiZr_{10}$ RHEAs was carried out and the oxidation resistance of $Al_{10}CrNbTiZr_{10}$ (Mo-0) examined, with results showing its potential for high-temperature applications such

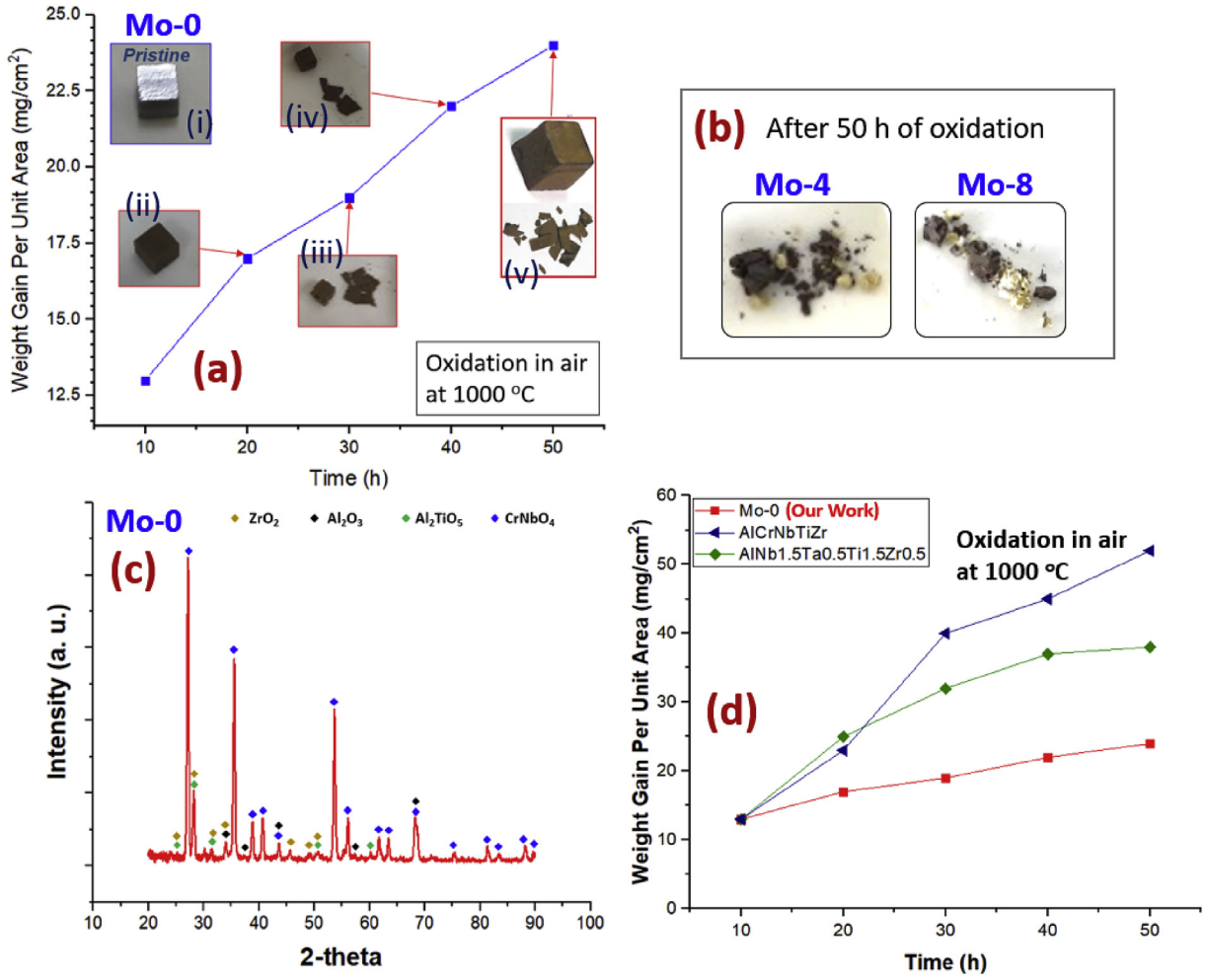


Fig. 5. (a) Weight gain in Mo-0 during prolong oxidation at 1000 °C in air, (b) Mo-4 and Mo-8 samples (disintegrated after 50 h of oxidation), (c) XRD analysis of oxidized Mo-0, and (d) comparison of weight gain with similar RHEAs oxidized under the same conditions.

as the components of power plants, turbines and automobile engines. However, the analysis of its mechanical properties is required and a point of further study.

4. Conclusion

The development and analysis of $Al_xTa_yV_z$-$Cr_{20}Mo_{20}Nb_{20}$-$Ti_{20}Zr_{10}$ ($Al_xTa_yV_z$-Q) and $Al_{10}CrMo_xNbTiZr_{10}$ was carried out. The $Al_xTa_yV_z$-Q samples were prepared by arc melting, and further homogenization of the microstructure was carried out by annealing at 1200 °C for 24 h in Ar.

The oxidation behavior of $Al_xTa_yV_z$-Q, when analyzed at 1000 °C for 1 h in air showed a weight gain ranging from 0.8 to 8.5 mg/cm^2. Al_{10}-Q exhibited the minimum weight gain (0.8 mg/cm^2); however, a negative slope in the TGA curve of Al_{10}-Q was observed, which is attributed to the formation of volatile oxides of Mo.

Therefore, the composition was modified to $Al_{10}CrMo_xNbTiZr_{10}$. The arc melting and homogenized $Al_{10}CrMo_xNbTiZr_{10}$ samples revealed a BCC microstructure with Al-, Zr-, and Cr-containing intermetallics. The TGA analysis of $Al_{10}CrNbTiZr_{10}$ for 1 h in air revealed a weight gain of only 1 mg/cm^2. The analysis of the oxide scale developed on Mo-0 after 1 h of oxidation, showed protective oxides of Al (Al_2O_3) and Cr (CrO_2 and $CrNbO_4$).

The prolonged oxidation behavior of $Al_{10}CrMo_xNbTiZr_{10}$ was analyzed by subjecting the samples to high temperature (1000 °C) for 50 h in air. The samples containing 4 at.% and 8 at.% Mo disintegrated after 10 h of oxidation, whereas $Al_{10}CrNbTiZr_{10}$ (Mo-0) remained intact and exhibited a weight gain of ~24 mg/cm^2 after 50 h. The formation of protective oxides ($CrNbO_4$, Al_2O_3, and $AlTiO_5$) enhanced the oxidation resistance of Mo-0 and suggested its potential for use in high-temperature applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Owais Ahmed Waseem: Investigation, Formal analysis, Writing - original draft. **Ho Jin Ryu:** Methodology, Supervision, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jallcom.2020.154427.

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